Class 2.

Tuesday, 18 August 2026 7:06 PM

Probability Space

(Ω, F, P)

$\Omega \rightarrow$ Sample space
: set of all possible outcomes of experiment

Eg Toss coin : $\Omega = \{\text{Heads}, \text{Tails}\}$ ✓

Eg Toss coin, observe speed : $\Omega = \mathbb{R}^+$ ✓

Eg Toss coin, till I see first heads ✓

$\Rightarrow \Omega = \{1, 2, 3, \ldots\}$

$F \rightarrow$ Collection of events

Event $\rightarrow$ Subset of sample space

Eg Event $E_1 \rightarrow$ I need ≥ 5 tosses

$E_1 = \{5, 6, 7, \ldots\} \subseteq \Omega$

$E_2 \rightarrow$ # tosses is odd $E_2 = \{1, 3, 5, 7, \ldots\} \subseteq \Omega$

$(\Omega, F) \rightarrow \mathbb{P} \rightarrow$ Probability measure

$\mathbb{P} : F \rightarrow [0, 1]$ Satisfy some prop.

$P(E) \in [0, 1]$

① $\mathbb{P}(\underbrace{\phi}_{\text{null set}}) = 0$, ② $\mathbb{P}(\Omega) = 1$

③ A, B which are disjoint, $A \cap B = \phi$

$\mathbb{P}(\underbrace{A \cup B}_{\substack{\text{elements which belong to} \\ \text{either A or B}}}) = P(A) + P(B)$

Eg Toss a coin, Observe side

$\Omega = \{H, T\}$ $F = \{\phi, \{H, T\}, \{H\}, \{T\}\}$

$\{H\}$: $\underbrace{P}_{p}$, $\{T\}$: $\underbrace{1-P}_{q}$

$P(\phi) = 0$, $P(\{H, T\}) = 1$.

$P(\{H\}) = P$, $P(\{T\})$

$A = \{H\}$, $B = \{T\}$. $A \cup B = \{H, T\} = \Omega$

$\underbrace{P(A \cup B)}_{1} = \underbrace{P(A)}_{P} + P(B)$

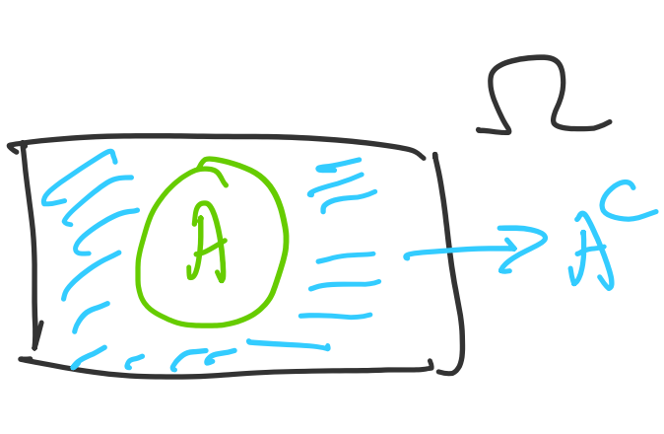
$\Omega = \{H, T\}$, $F = \{\phi, \{H, T\}, \{H\}, \{T\}\}$

$\phi \rightarrow 0$, $\{H, T\} \rightarrow 1$, $\{H\} \rightarrow P$, $\{T\} \rightarrow 1-P$

$P : F \rightarrow [0, 1]$

$(\Omega, F, \mathbb{P})$

① $A \underset{\text{belongs to}}{\in} F$, $A^C \in F$



$P(A^C) = 1 - P(A)$

or $P(A^C) + P(A) = 1$

A, A^C are disjoint. $A \cup A^C = \Omega$

$P(A) + P(A^C) = P(A \cup A^C) = P(\Omega) = 1$.

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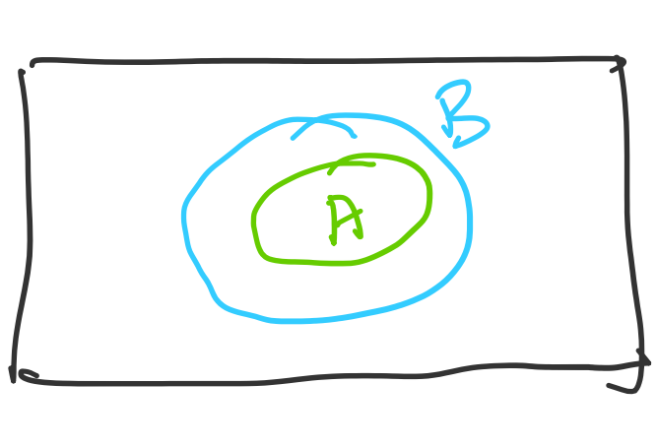
② A, B $A \underset{\text{is a subset of}}{\subseteq} B$



$\mathbb{P}(A) \leq \mathbb{P}(B)$

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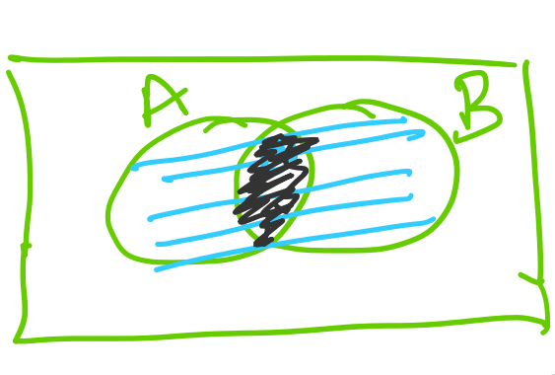
③ $P(A \cup B) = P(A) + P(B) - P(A \cap B)$



Eg User $\Omega = \{$Mobile - Purchase
Mobile - No Purchase
Desktop - Purchase
Desktop - No Purchase$\}$

$\Omega = \{MP, MN, DP, DN\}$

$MP \rightarrow .18$, $MN \rightarrow .42$, $DP \rightarrow .12$, $DN \rightarrow .28$

$P(\{MP, MN\}) = P(\text{User joins from mobile})$
$= P(\{MP\}) + P(\{MN\})$
$= .18 + .42 = .6$

$P(\{MP, MN, DP, DN\}) = .18 + .42 + .12 + .28 = 1$

$P(\text{User makes a purchase})$
$= P(\{MP, DP\}) = P(\{MP\}) + P(\{DP\})$ ($P(\{MP\}) \rightarrow .18$, $P(\{DP\}) \rightarrow .12$)
$= .3$

$A \rightarrow$ User joins from Mobile

$B \rightarrow$ User makes a purchase

$P(A) = .6$, $P(B) = .3$

$P(A \cap B) = P(\text{User joins from mobile \& makes a purchase})$
$= P(\{MP\}) = .18$

$A \cap B \subseteq A$, $A \cap B \subseteq B$

$A \rightarrow$ Mobile $= \{MP, MN\}$ ✓

$B \rightarrow$ Purchase $= \{MP, DP\}$ ✓

$A \cap B = \{MP\}$

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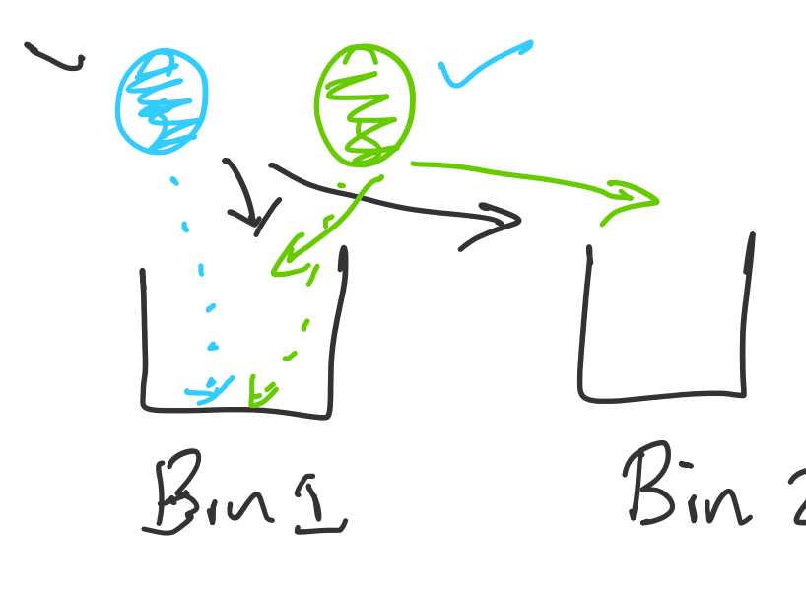
2 balls

2 bins



(1, 1) → 1/4

1, 2 → 1/4

2, 1 → 1/4

2, 2 → 1/4

Event E → There exists an empty bin

$P(E) = P(E_1 \cup E_2)$

$E_1 \rightarrow$ Event that Bin 1 is empty

$E_2 \rightarrow$ Event that Bin 2 is empty

$P(E_1 \cup E_2) = \underbrace{P(E_1)}_{\frac{1}{2} \times \frac{1}{2}} + \underbrace{P(E_2)}_{\frac{1}{2} \times \frac{1}{2}} - \underbrace{P(E_1 \cap E_2)}_{0}$

$= \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$

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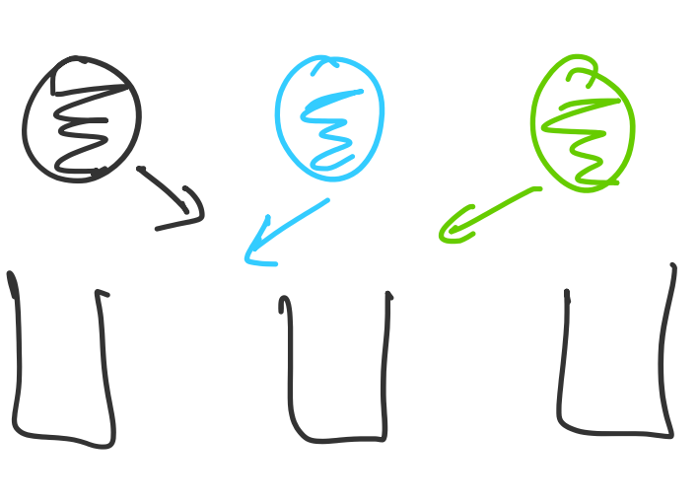
3 balls, 3 bins

$E =$ Exists an empty bin.



$E_1 \rightarrow$ Bin 1 is empty

$E_2 \rightarrow$ Bin 2 is empty

$E_3 \rightarrow$ Bin 3 is empty

$E = E_1 \cup E_2 \cup E_3$

$P(E) = P(E_1 \cup E_2 \cup E_3)$

Inclusion - Exclusion Principle

$P(A_1 \cup A_2 \cup A_3) = P(A_1) + P(A_2) + P(A_3)$
$- [P(A_1 \cap A_2) + P(A_2 \cap A_3) + P(A_1 \cap A_3)]$
$+ P(A_1 \cap A_2 \cap A_3)$

$P(E_1) = P(\text{Bin 1 is empty}) = \frac{2}{3} \times \frac{2}{3} \times \frac{2}{3}$

$\frac{2}{3} \leftarrow$ Black ball not in Bin 1 & $\frac{2}{3}$ Blue Ball not in Bin 1 & $\frac{2}{3}$

$P(E_1) = P(E_2) = P(E_3) = \frac{8}{27}$

$P(E_1 \cap E_2) = P(\text{Bin 1 \& Bin 2 are empty})$
$= \frac{1}{3} \times \frac{1}{3} \times \frac{1}{3} = \frac{1}{27}$

$P(E_1 \cap E_3)$ ''

$P(E_2 \cap E_3)$ ''

$P(E_1 \cap E_2 \cap E_3) = P(\text{Bin 1 \& Bin 2 \& Bin 3 are empty})$
$= 0$

$P(\text{exists an empty Bin})$

$= P(E_1 \cup E_2 \cup E_3)$

$= \frac{8}{27} + \frac{8}{27} + \frac{8}{27} - [\frac{1}{27} + \frac{1}{27} + \frac{1}{27}] + 0$ $= \frac{24}{27} - \frac{3}{27} = \frac{21}{27}$

A = Exactly 1 bin being empty

3 bins.

A^C

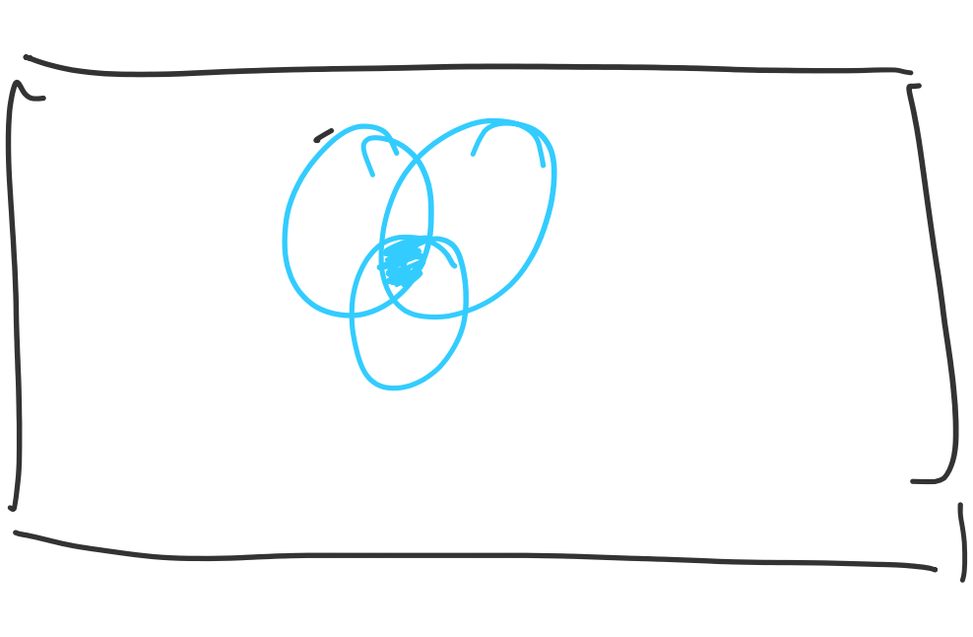
Bin 1, Bin 2, 3 are not empty

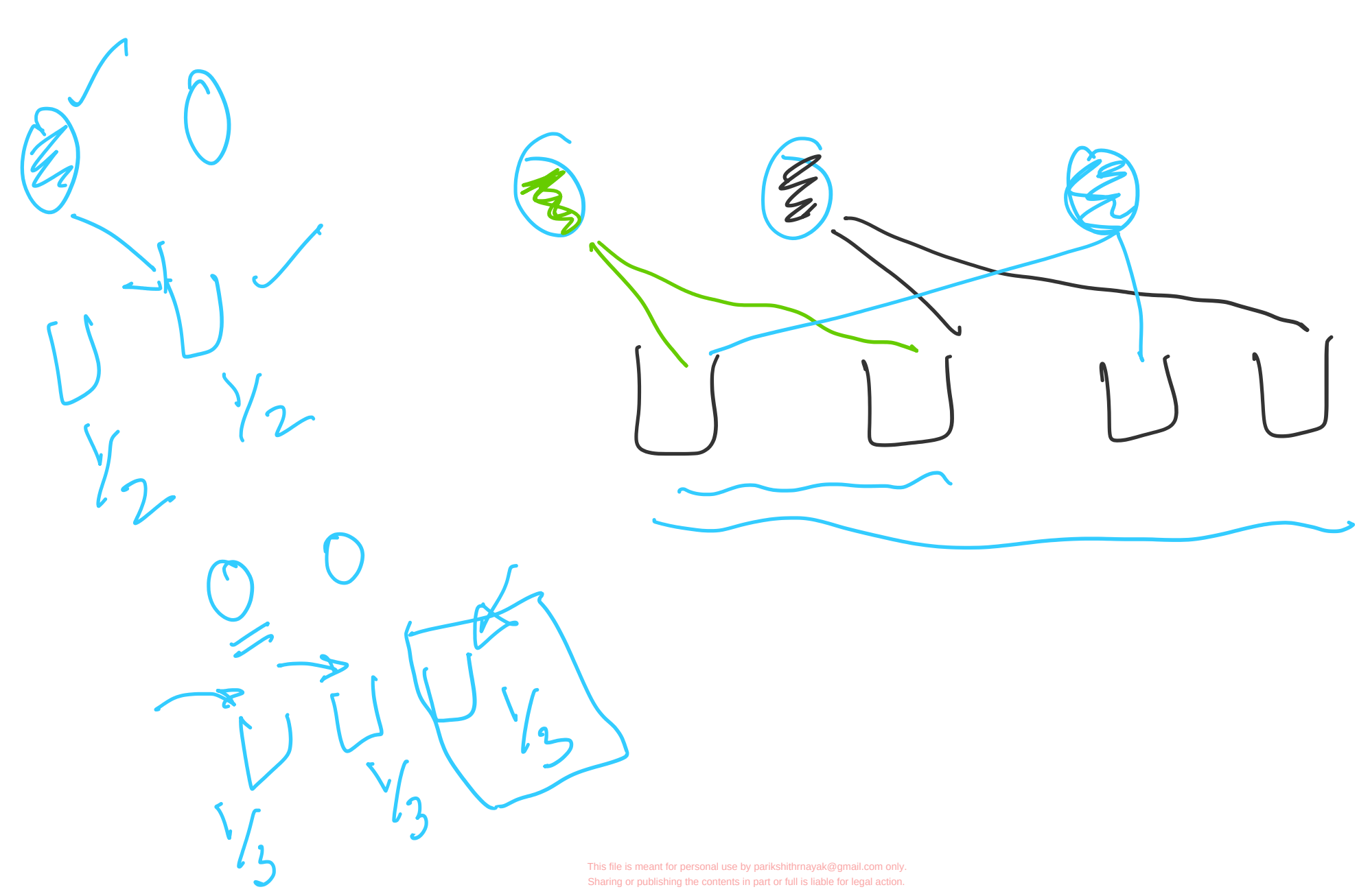
Bin 2

$A^C = \{\text{all bins not empty}\} \cup \{1, 2 \text{ empty}\} \cup \{2, 3 \text{ empty}\} \cup \{1, 3 \text{ empty}\}$

$P(A^C)$

$P(A) = 1 - P(A^C)$





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